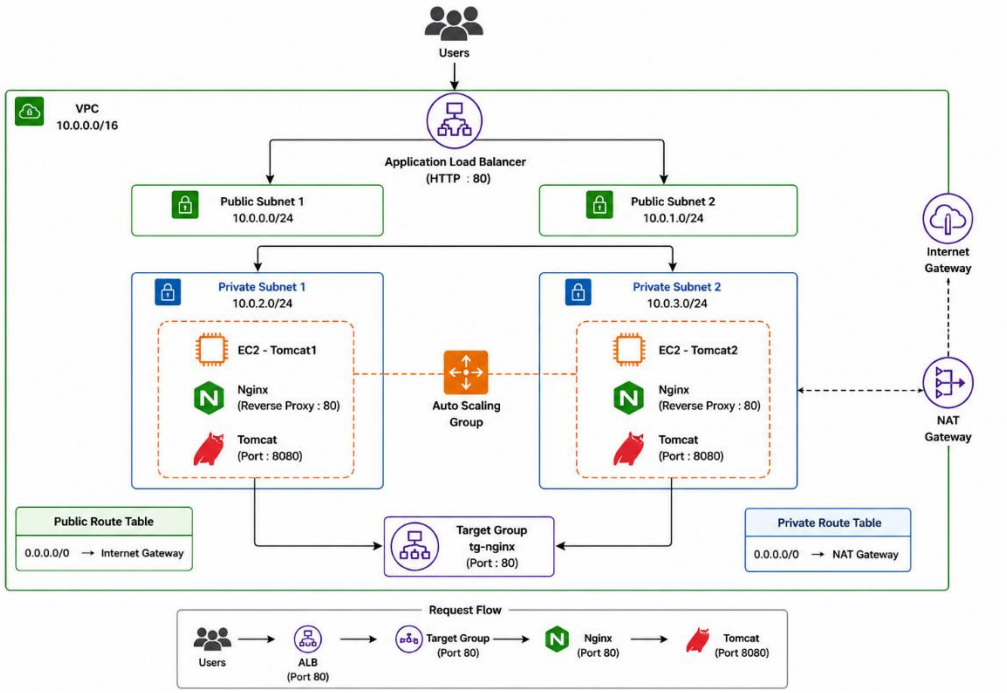
INFRASTRUCTURE-1



1. create a vpc-(my-vpc)-(10.0.0.0/16)
2. create a subnet
    1. public-subnet-1 (az-1a) (10.0.0.0/24)
    2. public-subnet-2 (az-1b) (10.0.1.0/24)
    3. private-subnet-3 (az-1a) (10.0.2.0/24)
    4. private-subnet-4 (az-1b) (10.0.3.0/24)
3. create a internet-gateway and attach to vpc (my-vpc)
4. create a nat-gateway regional (my-vpc)
5. create a route-table
    1. public-route-table
        a) associate-public-subnet 1&2
        b) edit routes ---> 0.0.0.0/0 -> internet-gatway
    2. private-route-table
        a) associate-private-subnet 3&4
        b) edit routes ---> 0.0.0.0/0 -> nat-gatway

6. create a launch-template
    1. name ----> mylt
    2. tags ------> Name-tomcat
    3. instance_type ----> t3.micro
    4. os_machine --> amazon-linux
    5. key_pair -------> ubuntu
    6. create a security group
        a) name ----> my-sg

b) vpc -----> my-vpc
c) permission --> ssh/http/https/All-TCP

7. create a template

7. create a autoscaling group
    1. name ----> ASG
    2. lt ------> mylt
    3. subnets -> private 3&4
    4. create a autoscaling group

8. create a bastion-host-server
    1. access the bastion-host-server --> get the access-tomcat1
    2. sudo dnf update -y
    3. sudo dnf install java-17-amazon-corretto-devel -y
    4. wget https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.118/bin/apache-tomcat-9.0.118.tar.gz
    5. tar -xzf apache-tomcat-9.0.118.tar.gz
    6. cd apache-tomcat-9.0.118
    7. cd bin
    8. chmod +x catalina.sh
    9. ./catalina.sh start
    10. curl localhost:8080

9. access tomcat-2 on bastion-host-server
    1. sudo dnf update -y
    2. sudo dnf install java-17-amazon-corretto-devel -y
    3. wget https://dlcdn.apache.org/tomcat/tomcat-10/v10.1.55/bin/apache-tomcat-10.1.55.tar.gz
    4. tar -xzf apache-tomcat-10.1.55.tar.gz
    5. cd apache-tomcat-10.1.55
    6. cd bin
    7. chmod +x catalina.sh
    8. ./catalina.sh start
    9. curl localhost:8080

10. create a target-group
    1. name ----> tg
    2. vpc -------> my-vpc
    3. register -> tomcat1 & tomcat2
    4. 8080 -----> include as pending below
    5. create a target group

11. create a load-balancer

1. name ----> ALB
2. vpc -----> my-vpc
3. az ------> public-subnet 1&2
4. tg ------> tg
5. create a load-balancer

12.Copy the DNS & hit it on web
1. http://alb-869679179.ap-south-1.elb.amazonaws.com:8080/

TO HIDE THE PORT:8080 (IMPLEMENTING NGINX-REVERSE-PROXY)

13.Access tomcat-1 on bastion-host-server
1. sudo dnf update -y
2. sudo dnf install nginx -y
3. sudo systemctl enable nginx
4. sudo systemctl start nginx
5. sudo systemctl status nginx
6. vim /etc/nginx/nginx.conf

```
server {
   listen 80;
   listen [::]:80;

   server_name _;

   location / {
      proxy_pass http://localhost:8080;

      proxy_set_header Host $host;
      proxy_set_header X-Real-IP $remote_addr;
      proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
   }

   error_page 404 /404.html;
   location = /404.html {
   }

   error_page 500 502 503 504 /50x.html;
   location = /50x.html {
   }
}
```

7. nginx -t
8. sudo systemctl restart nginx
9. curl localhost

14.Same steps for the tomcat-2

15.create a new target-group

1. name ---> tg-nginx
2. vpc ------> my-vpc
3. register---> tomcat 1&2
4. port ---> 80 ---> include as a pending below
5. register targets

16.configure load-balancer

1. click on listener ---> add listener
2. http ----> 80
3. forward to tg ---> tg-nginx
4. create

17.copy the dns name

1. http://alb-869679179.ap-south-1.elb.amazonaws.com
